

Chapter 2

Evolution of University-Industry Relationships for Driving Innovation



Tomas Coates Ulrichsen

2.1 Introduction

The past 50 years have seen dramatic changes in the innovation endeavors of many nations. We have witnessed changes to the nature, complexity, and focus of innovation processes encompassing organizations involved in delivery, government agencies that share priorities and provide support, and institutions that enable relationships. Alongside these changes, we have seen major changes to the scale, nature, and importance of the interactions formed between universities and companies to drive innovation. Today, governments around the world are positioning universities and their partnerships with companies, government labs, philanthropy, investors, and others, as critically important for their efforts to drive innovation and economic growth, and for tackling major global challenges.

This chapter explores the evolution of these University-Industry Relationships (UIRs) and the implications of these changes for the different key stakeholders involved. It first looks at the changing nature of innovation and the systems within which these interactions take place before looking at how UIRs have evolved within this context and where we might be heading. It then explores what we know about the key factors that shape their development. It concludes by considering the implications of the evolving importance, scale, and nature of UIRs for key stakeholders involved in the process.

T. C. Ulrichsen (✉)

Policy Evidence Unit for University Commercialisation and Innovation (UCI), University of Cambridge, Cambridge, UK

e-mail: tc267@cam.ac.uk

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

R. Hall, A. Boccanfuso (eds.), *University-Industry Collaboration*, International Series in Operations Research & Management Science 369, https://doi.org/10.1007/978-3-031-94913-5_2

2.2 Literature on an Evolving Innovation Landscape

The interactions between universities and companies that form with an ambition to drive innovation take place within the context of a wider innovation system which itself is constantly evolving. At the heart of these systems is an innovation process seeking to develop, transform, and deploy knowledge to produce innovations for economic and societal gain (Edquist, 1997). Different organizations contribute different inputs and activities to the process at different points along the journey, and both market and nonmarket interactions between them are important for ensuring effective knowledge flow. The innovation process and the actions and behaviors of organizations are shaped by both formal and informal “rules of the game,” including national policy and legal frameworks, regulations, markets (financial, labor, trade etc.), as well as cultural norms and values.

This section considers how the system has evolved through the twentieth century and into the 2000s due to changing: (1) nature of the innovation process; (2) national policy and legal frameworks governing science, technology, and innovation; and (3) roles and positions of universities and companies.

2.2.1 *The Changing Nature of Innovation*

The twentieth century saw knowledge becoming an increasingly important factor of production alongside physical and human capital in driving economic growth, with economic opportunities underpinned by major technological advances becoming a defining feature (Arora et al., 2020; Kuznets, 1971), and technology and knowledge intensive industries taking center stage in many advanced economies (Foray, 2004). Economies experienced rapid technological change, shorter product life cycles, growing complexity of products and services that were increasingly built on advances at the scientific and technological frontiers, and increasingly intense global competition (Arora & Gambardella, 1994; Bettis & Hitt, 1995; Santoro & Chakrabarti, 2001). As a result, the entrepreneurial dynamism of an economy, the ability to produce knowledge and generate and commercialize inventions emerging from research, and the capacity and capability to turn these inventions into successful innovation, became important drivers of long-run economic growth and competitiveness, growth and survival of companies (Audretsch, 2014; Bettis & Hitt, 1995).

Alongside this, we witnessed significant shifts and developments in our understanding of how innovation happens, which in turn shaped our approaches to framing key issues and managing the process (Galanakis, 2006; Tidd, 2006). Early models saw innovation conceived as a linear process, following a sequence of functional activities from research to development, production and marketing. Signals for innovation emerged either through “*technology push*” (dominant paradigm in the 1950s), where major scientific advances create the opportunities and drive for

new products and services, or “*demand pull*,” whereby the signals for innovation are driven by market need (common in the 1960s).

The limitations of these approaches became evident in the 1970s, with recognition that innovation processes were “neither smooth nor linear, nor often well behaved” (Kline & Rosenberg, 1986, p. 285). This gave rise to the *coupling models* of Kline and Rosenberg (1986), which saw much greater emphasis on the feedback and interactions between different stages of the process. Much more attention was given to knowledge flows between the stages of the process and the wider research base. Advances in new product design and development theory then saw innovation processes reconceived as parallel, rather than sequential processes often simultaneously engaging multiple functions of the firm.

The rise of new digital tools, rapid prototyping and the proliferation of information and communications in the 1990s led to a growing focus on innovation as a process of *systems integration and networked innovation*, with inputs from both the scientific community and from markets shaping decisions (Tidd, 2006). These models began to explicitly recognize the importance of suppliers and customers and of alliances in the innovation process, as well as the role of knowledge, creativity, and learning as sources of innovation.

The emergence of *systems of innovation* theory during the 1990s led to innovation being explicitly conceived as a collective activity underpinned by learning processes that involved networks of actors interacting with each other through both market and nonmarket links, and within a specific set of institutions that shape the “rules of the game” (Lundvall et al., 2002). These activities can be economic (e.g., financial markets), legal (e.g., IP regimes), political and governmental (e.g., government policies and political priorities), and social (e.g., cultural attitudes towards innovation). Despite often being presented as different generations of innovation models, in reality they likely coexist in the real world, with different approaches deployed under different circumstances (Caraça et al., 2009).

In recent years we have seen the rise of additional drivers of innovation, which, in turn, influence the nature of innovation. These include, not least: rising geopolitical tensions and national security concerns; recognition of the growing number of global grand challenges faced by nations and societies around the world that need to be tackled urgently; the growing importance of social progress, equity, and environmental sustainability; and the acceleration of digital transformation of industry and societies along with the rise of increasingly sophisticated artificial intelligence tools.

2.2.2 *Changing National Policy Frameworks for Innovation*

National policy frameworks for science, technology, and innovation have co-evolved with the changing nature of innovation. These policy frameworks shape the types of public funding programs available to support innovation. They also help to set wider incentives and expectations on the innovation system and play a part in

influencing why and how universities and companies come together to advance innovations.

National science and innovation policy frameworks in the USA, Europe, UK, and elsewhere have undergone significant paradigmatic shifts since the 1940s, with three broad phases identified (Lindner et al., 2024; Schot & Steinmueller, 2018): (1) science for growth and addressing market failures; (2) systems of innovation and systems upgrading; and most recently (3) a growing focus on transformative change and tackling societal challenges.

The first of these phases—science for growth—emerged during the 1940s and was born out of the experiences of the Second World War and the need to confront postwar futures including the rise of the Cold War and the need for economic renewal. Massive expansions in scientific investments by governments in Research and Development (R&D)—increasingly in research universities—had led to major discoveries and innovations, and demonstrated to policymakers the potential for significant returns (Arora et al., 2020; Lindner et al., 2024). Informed by the linear model of innovation and economic theories that emphasized the importance of science and technology in driving growth (not least Arrow, 1962; Solow, 1957), post-war science and technology policies in developed economies emphasized the role of governments in strengthening their science systems and funding basic knowledge that could lead to discoveries and find their way into innovations either via the market mechanism or via direct demand from the state (particularly for military innovations) (Lindner et al., 2024).

Science and technology policies during this period also sought to promote technology transfer from the public to private sector (including through the strengthening of Intellectual Property (IP) rights—for example with the 1980 Bayh-Dole Act in the USA), incentives to support research in the private sector (e.g., through R&D tax credits and public procurement), the importance of venture capital, and efforts to tackle the anticompetitive effects of market dominance (e.g., through antitrust regulations). Research universities were also positioned as having an important role to play in delivering a skilled workforce able to apply new knowledge to tackle societal problems (Conn et al., 2021).

The 1980s and 1990s saw the emergence of a new generation of science and innovation policies that were focused more on the development and upgrading of *systems of innovation*. This shift was driven not least by a growing questioning of the science for growth policy paradigm and a perceived failure to deliver the expected returns across the world, along with the recognition that (Freeman, 1982; Lindner et al., 2024; Schot & Steinmueller, 2018):

- Innovation was an interactive process involving multiple actors and appearing more complex than the linear model would imply
- There is an important interplay between science and innovation with society and the role of not just economic but also sociopolitical institutions in shaping innovation behaviors
- Tacit knowledge was important in shaping knowledge flows between organizations and across borders

- The ability of companies and nations to absorb and exploit knowledge was important in addition to the ability of the system to produce it
- The capability to learn and be creative is important in driving innovation
- Entrepreneurship was important in modern innovation-driven economies.

Compared to the science for growth paradigm, the systems of innovation approach to policy highlights a much wider set of failures in different parts of the system that act to hamper the overall development, deployment, and diffusion of innovations for socioeconomic gain (Kelleher & Ulrichsen, 2022; Weber & Rohracher, 2012). These include failures in the:

- Sociopolitical, legal, and economic institutions (including IP regimes as well as cultures towards innovation) that influence innovation actions and behaviors
- Networks and linkages between actors that enable that knowledge to flow
- Lock-in whereby the system becomes excessively focus on existing technologies and faces very large barriers to adopting and diffusing new technologies even if they may deliver significant benefits
- Scientific, physical, and network infrastructure that hampers innovation activities
- Capabilities and resources of actors that make it harder to access and (co-) develop new knowledge and acquire and translate it into new innovations delivering value, and that make it harder to adapt, reconfigure, and learn as opportunities and challenges change
- Imbalances between exploration and exploitation within the system that result in too little (too much) focus on creating new technological options and too little (too much) emphasis on translating and scaling new ideas into innovations.

Crucially, alleviating these failures often requires actions from a range of policy domains to strengthen not just the supply side of the system but also the demand side, their ability to interact, and the sets of institutions that shape behaviors.

The systems of innovation policy paradigm positions universities as key actors alongside startups, corporates, investors, and others. It emphasizes the importance not just of technology transfer, but also of the ability of these organizations to form much closer and more interactive *partnerships* to better integrate knowledge production with its translation and deployment. Both formal and informal interaction mechanisms are seen as important in facilitating two-way flows of codified and tacit knowledge necessary for innovation, and companies and other knowledge users also require incentives, capabilities, and resources to engage and benefit from knowledge generated within the university system. There is also a recognition of the value of universities and their partners in building up and leveraging not just their R&D capabilities but also a wider set of assets and resources to support the innovation endeavor, including not least facilities and equipment (e.g., to support prototyping, demonstration and testing of emerging technologies, and problem solving), education and training, and professional networks and alliances. Furthermore, it highlights the need for efforts and investments to better align cultures and incentives within the research and innovation base towards working better together to deliver innovation, to strengthen the entrepreneurial activities and capabilities of

universities, and to create spaces for bottom-up experimentation and learning within the system.

Systems innovation policy dominates in many countries today, including the UK, USA, and much of Europe. However, in the last few years, a new paradigm for science, technology, and innovation policy focused on a paradigm of **transformative change** has emerged. Although still in its infancy, transformative change is motivated by the view that the systems of innovation approach is too passive and reactive, given the major societal, economic, and environmental challenges faced by societies around the world, such as climate change, biodiversity loss, aging populations, health challenges (e.g., antimicrobial resistance and dementia), and resource scarcity (Chaminade et al., 2018). Scholars argue that many of these challenges are inherently complex “wicked” problems, and tackling them requires policy to become much more proactive at delivering transformative change (Chaminade et al., 2018; Lindner et al., 2024; Mazzucato, 2018; Schot & Steinmueller, 2018; Weber & Rohracher, 2012). It needs to be recognized that there is no single best pathway to tackle wicked problems; rather, disruptive change will see many actors negotiating alternative pathways that have the potential to deliver system change. Experimentation throughout the system, the accumulation of experience by a variety of actors, and facilitating pathways to scale up experiments, are central to the paradigm (Schot & Steinmueller, 2018).

The transformative change paradigm has also shifted the role of government. For example, failures articulated by the systems of innovation approach highlight transformational failures, which need to be addressed. *Directionality* failures emerge due to difficulties in setting shared visions, the lack of means for making social choices among alternative pathways for development, and challenges in setting collective priorities given the often competing interests of different groups. *Policy coordination* failures exist due to difficulties in coordinating policy action across multiple areas of government (fiscal policy, industrial policy, social policy etc.) as well as at different levels of government from local to regional to national and international (Schot & Steinmueller, 2018). Improved coordination is crucial to overcome these failures. To deploy and scale many solutions to grand challenges requires not just technological solutions but also changes in behaviors and the structure and function of societal, economic, and industrial systems (Mazzucato, 2018; Schot & Steinmueller, 2018).

The ability to deliver transformative change is also hampered by *demand articulation* failures resulting from difficulties in understanding and anticipating knowledge and innovation user needs and the challenges faced by these users in deploying and adopting solutions. *Reflexivity* failures also emerge from the long-term nature of transformative change and the uncertainties and complexities surrounding the innovation and change required. This complexity, scholars argue, requires continuous monitoring of progress towards transformation goals, the ability to host reflexive discourses, strategies for adaptation, portfolio approaches, self-governance arrangements that allow for the involvement of diverse groups and the distributed nature of decision-making and intelligence, and deep learning that allows groups to critically reflect on, and question, their own deeply-rooted preferences and routines that drive

people towards optimizing the status quo rather than actions and opportunities that deliver transformative change (Schot & Steinmueller, 2018; Weber & Rohracher, 2012).

This approach, while still in its infancy with few examples of widespread implementation (Lindner et al., 2024), suggests additional roles for universities working in partnership with companies, investors, governments, and others (Kelleher & Ulrichsen, 2022). This includes, for example, working much more actively together and with governments to inform the priorities, focus and direction of innovation policies and strategies; enhancing existing roles as neutral convenors to provide spaces for dialogues, bringing together actors from across the system to gather intelligence on directions-of-travel and progress towards goals; and helping to understand the alternative pathways emerging and different system-level opportunities and consequences of different courses of action.

2.2.3 *Rise and Fall of Corporate R&D Labs*

The growing importance of innovation in economic growth and competitiveness also saw knowledge producing organizations take center stage in innovation and growth strategies (Audretsch, 2014). Alongside the evolution in *how* innovation happens, we witnessed dramatic shifts during the twentieth century in *who* was central to the process of knowledge production, translation, and innovation (Arora et al., 2020).

The early part of the twentieth century saw individual inventors and entrepreneurs give way to the rise and proliferation of large corporate R&D labs as the primary source of invention in the United States (Arora et al., 2020). Their core objective was to deliver research to support the existing product portfolios and unlock major new commercial opportunities for their company. Their investments in research reflected their belief that scientific knowledge—even more fundamental knowledge—was important for guiding product development (Arora et al., 2020). They delivered major advances in science and engineering—including high-quality scientifically complex research that, although aimed at solving industrial challenges, rivaled or even exceeded that of top research universities and resulted in Nobel prizes that unlocked new opportunities in areas from chemicals, to automotive and aerospace, pharmaceuticals, semiconductors, information and communications technologies, and many others (Arora et al., 2020). Their successes led to efforts to routinize innovation and shift away from the previously uncertain supply of inventions to drive new economic opportunities. During this “golden era” of the corporate R&D lab, they employed large numbers of highly capable scientists, often recruited from, and trained by, universities. Their links to universities were also strong and important in their successes, as was access to significant funding by the federal government (Arora et al., 2020).

By the 1980s, however, the corporate R&D lab had fallen out of favor with many large corporations. Questions were asked about their ability to deliver the research

needed to drive the innovation needed to underpin competitiveness in an era characterized by growing product market competition, shorter technology lifecycles, more complex technologies and increasingly demanding investors. R&D began to shift away from large central R&D functions to R&D tied to products or divisions or at least imposition of much greater coordination of the research agenda between central R&D and product and division managers (Gerybadze & Reger, 1999). It also increasingly moved to more open and collaborative models that sought to leverage much larger and more diverse knowledge bases external to the company (Arora et al., 2020; Chesbrough, 2003; Laursen & Salter, 2006; Tasse, 2007).

2.2.4 Evolving University System

As the nature, delivery, and priorities for innovation have evolved, university systems have co-evolved, particularly in the USA, UK, and Europe. Over a timespan of several hundred years, different types of universities have emerged, responding to different pressures and opportunities. Perhaps most talked about is the classical research university modeled on the Humboldt University of Berlin in the 1800s, with fundamental research driven by intellectual curiosity at its core, funded generously by the state, with research strongly integrated with teaching. By the end of the nineteenth century it had become one of the world's leading research institutions and its model had spread widely (Martin, 2012). It also had a major influence on government approaches to funding universities, in Europe in particular.

Alongside the classical research university, other types of universities were emerging and thriving. Martin (2012) highlights examples, such as University College London, King's College London, and the Universities of Birmingham, Sheffield, and Manchester across the UK. These universities were all dedicated to supporting the needs of a modern industrial society and worked closely with local and national industry to deliver their ambitions. Technical higher education institutions and institutes of technology also emerged in many countries with a strong focus on professional training and meeting the technology needs of industry. These included, for example, MIT, Georgia Tech, and Caltech in the United States; Imperial College of Science Technology and Medicine in the UK; KTH Royal Institute of Technology Stockholm in Sweden; and the Technische Hochschulen in Germany (Martin, 2012). In the USA, the Morrill Act of 1862 and the subsequent Morrill Act of 1890 provided land and federal funding for US states to endow colleges to significantly expand higher education opportunities and deliver research and technical support for local industries (in particular agricultural and mechanical sectors) (Arora et al., 2020; Martin, 2012). Amongst other things, these colleges provided agricultural extension services and experimental field stations to help diffuse the latest technologies across their states. From these origins, the universities evolved to develop sets of institutional mechanisms for commercializing science and technology to support innovation in their local economies and nationally, which

some argue were an important driver behind US agriculture becoming the most productive in the world (Audretsch, 2014).

Given the focus of universities and colleges during the 1800s, and many companies not having in-house R&D capabilities, industry across Europe and the USA frequently turned to the university system to solve their technical challenges and ensure they had a well-trained workforce (Martin, 2012).

Research universities in particular grew dramatically in the postwar period, benefiting from the significant expansions in government funding for research (Arora et al., 2020; Martin, 2012; Youtie & Shapira, 2008). With the decreased reliance on industry funding for research that had dominated in the first half of the twentieth century, research universities found themselves increasingly decoupled from the specific innovation needs of industry and became more focused on faculty-led research and building up stocks of human capital through their education programs. The 1960s and 1970s also saw growing suspicion within university faculties on both sides of the Atlantic of working closely with industry to address industry-led innovation opportunities, in some cases leading to withdrawal from such activity to focus on more academic-led discovery (Arora et al., 2020; Youtie & Shapira, 2008).

This began to change in the last few decades of the twentieth century and into the 2000s. University systems continued to co-evolve with the wider system, facing pressures and opportunities not least from decline of corporate R&D labs, the fall of the Berlin wall and the end of the Cold War shifting demand for R&D from military towards more civil-focused applications, growing global competition, and increasing pressures from governments to leverage their research bases to drive national competitiveness and economic growth. We witnessed a renewed appreciation—within governments and university faculties—of the importance of research universities in pursuing not just basic research but also in being able to orient discovery research towards use in society (often termed “use-inspired basic research”) and deliver more applied research able to be more readily translated into applications (Arora et al., 2020; Stokes, 1997).

Increased interest in the economic roles of universities and in the ability of universities to commercialize their research led governments to introduce new legislation, policies and funding programs aimed at strengthening the incentives and resources available to universities to commercialize research and build links with industry.

In the USA, these events famously resulted in the passage in 1980 of the Bayh-Dole Patent and Trademark Amendments Act, recognizing that few inventions emerging from the significant increases in federally funded research since the Second World War were making it to market. The Act gave universities and the scientists delivering federally funded research the rights to commercialize, and profit from, any discoveries emerging from this research (Shane, 2004). A key aim was to reduce uncertainty and incentivize universities to actively engage in this activity. Universities responded by investing in their technology transfer infrastructure and support for their scientists seeking to commercialize research. Coupled with wider developments in the US institutional, legal, and funding environments—including the significant expansion of the US government’s Small Business

Innovation Research (SBIR) Program in 1982 that supports the de-risking of technologies through public procurement channels, and growing access to venture capital—universities saw their licensing and spinout activity grow significantly in the 1980s and 1990s (Arora et al., 2020; Shane, 2004).

The 1970s and 1980s also saw the introduction of targeted funding programs by the US National Science Foundation (NSF) aimed at strengthening the links and partnership between universities and industry to drive technology development and bolster US competitiveness. These included the Engineering Research Centers (ERC) program, which focused on driving interdisciplinary and convergent research, education and innovation around engineered systems; and the Industry-University Cooperative Research Centers (IUCRC) program to foster long-term partnerships between universities, industry and government to support research, technology transfer, and workforce development in areas of mutual interest.

Across the Atlantic, the 1980s and 1990s saw the UK government make a number of major changes aimed at strengthening the partnership between universities and industry, creating stronger incentives for universities and academics to engage in the commercialization of research (Agar, 2019; British Academy, 2019). This followed a heavy emphasis through the 1970s on basic research as the driver of technological progress and innovation. Changes included the removal in 1983 of the government-owned British Technology Group's (formerly the National Research Development Corporation) monopoly over university IP emerging from research funded by the UK's Research Councils. Consequently, universities attained responsibility to commercialize resulting research.

The early 1980s also saw the UK launch the Alvey program to boost the nation's technological and industrial competitiveness in the information technology space. Jointly funded by government and industry, the program focused on close collaboration between academia and industry. Other developments included allowing university scientists to charge for consultancy and to exploit their research for commercial gain; tax incentives for investments in startup companies that helped to attract venture capital into the system; and growing expectations that research activities should produce a commercial return to the UK economy (Agar, 2019; British Academy, 2019).

Another key turning point in the UK was the 1993 publication of a new government strategy for science, engineering, and technology (*Realising Our Potential*). This was the first major review of science, engineering, and technology policy since the early 1970s, and set expectations of a "closer partnership and better diffusion of ideas between the science and engineering communities, industry, the financial sector and Government are needed as part of the crucial effort to improve our national competitiveness and quality of life" (HMSO, 1993, p. 8). This review laid crucial groundwork for the introduction in 1999 of a dedicated funding stream for universities to build up their capacity and capabilities to engage with industry and others to drive increased socioeconomic gains for the UK.

The calls for universities to become more actively engaged in technology transfer and partnerships with industry were also gathering pace in China in the 1980s and 1990s. China introduced a series of legislative measures designed to encourage

invention and its application to economic purposes, and in 1993, introduced the Scientific and Technological Progress Law, its equivalent of the US Bayh-Dole Act (Chen et al., 2016). The law required university employees to transfer the IP rights of inventions to their university. Subsequent changes later that decade included allowing universities to establish their own firms and allowing researchers to take sabbaticals to engage in technology transfer and start new ventures while also clarifying university responsibilities in protecting IP.

Entering the 2000s, universities were becoming increasingly important innovation partners for companies and were facing growing pressures—from governments and industry alike—to become more actively involved in delivering economic benefits from their research. Many universities were also under increasing pressure to diversify their income base. Within this context, universities continued to evolve and adopt new roles, becoming “more deeply embedded in innovation systems, seeking to actively foster interactions and spillovers to link research with application and commercialization, and taking on roles of catalyzing and animating economic and social development” (Youtie & Shapira, 2008, p. 1189).

This period saw the rise of a more formal and strategically important innovation and economic development mission, sometimes referred to as the “third mission” for universities, alongside research and teaching (Ulrichsen, 2024). In countries like the UK, the third mission has focused much more broadly than promoting technology transfer. It also covers the full range of ways through which knowledge could be exchanged and diffused to drive innovation.

In the UK, the rise of the innovation and economic development mission for universities was greatly facilitated by rapidly growing attention from policymakers on the need to more closely integrate universities with the wider economy to ensure the country could benefit from the ideas and knowledge generated (Ulrichsen, 2024). Within the UK, in 1999 the English universities funding body introduced a dedicated funding program for universities in England to support the development of their third mission.¹ It provided, for the first time, dedicated and flexible resources to universities *as institutions* to invest in building up their institution-wide capacities and capabilities to engage with external partners to exchange knowledge to drive innovation and economic development based on local needs, opportunities, and strategic priorities. These programs grew from approximately £50 million per year in the early 2000s (in 2023 prices) to £150 million in 2008–09 and over £300 million per year in 2023–24.

Crucially, over this period, the main funding programs for knowledge exchange for English universities switched from project-based competitions to more predictable formula-driven funding allocations to provide long-term stability to the higher education sector. Evaluations have shown that this long-term stability and flexibility have been incredibly valuable in enabling universities to significantly expand their engagements with industry and other partners to exchange and transfer knowledge

¹Funding for the third mission universities as organizations is devolved within the UK to each of the four nations, England, Scotland, Wales and Northern Ireland. The discussion here refers specifically to developments in England.

to drive innovation and economic development (PACEC/CBR, 2009; Ulrichsen, 2015). It has enabled universities to invest in a *system* of support for knowledge exchange, with developments and investments in (Ulrichsen, 2024) in the following changes:

- New leadership and strategies
- Improved incentives and efforts to change culture
- Creating dedicated support units and programs
- The expansion of challenge-led or user-focused centers and institutes bringing together research, translation, and knowledge exchange
- Developments of university campuses to foster the collocation of industry, the translation of research, and support innovation in the local economy
- The introduction of internal funds to support research translation and engagements with industry
- Improving and streamlining the processes and approaches to delivering knowledge exchange
- Providing training and other support—including Professors of Practice, mentors and experts-in-residence aimed at building the capabilities of academics and others to work more effectively with external partners

Expanded roles of universities, such as these, working closely with industry and other partners to drive innovation and economic development, are also evident in the USA, as evidenced, for example, with the passage of the recent Chips and Science Act of 2022 and the introduction of the new Technology, Innovation and Partnerships Directorate within the NSF to advance use-inspired and translational research. Amongst other things, these developments place a well-functioning partnership between universities, industry, investors, and others at the center of efforts to strengthen the translation of research into technologies, products, and services to ensure the continued global leadership and competitiveness of the USA in strategically important technologies and industries. The Regional Innovation Engines program introduced by the Act to ensure that more parts of America can contribute to, and benefit from, the national innovation endeavor, also emphasized the important role of universities working in close partnership with others in driving regional innovation and competitiveness.

The changes in university roles have helped to more fully embed a broader set of roles for universities in the twenty-first century that extend well beyond generating breakthrough technological innovations to delivering advances in a wide range of innovation areas including production processes, business models, and working practices (Ulrichsen & Kelleher, 2022). The changes also deliver services that leverage university expertise and infrastructure that help partners deliver innovation activities. Universities are also increasingly working with local partners to strengthen the conditions, resources, and capabilities available within their local innovation systems to better enable companies to innovate. In addition, they are helping introduce emerging ideas and technologies into the market. These benefits are summarized in Fig. 2.1. Looking to the future, we are starting to see universities evolve further to take on greater roles in tackling complex grand challenges that often

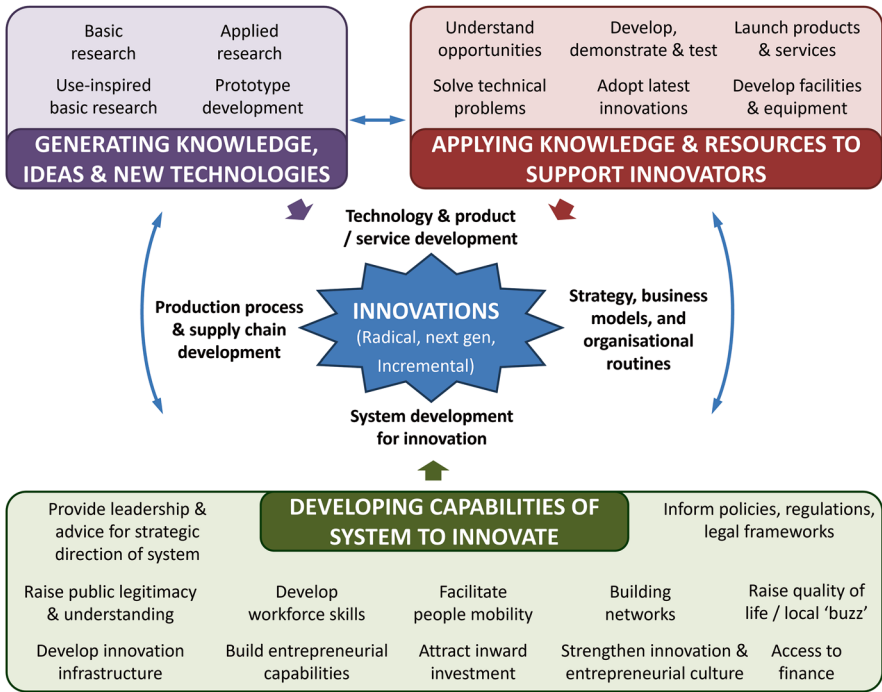


Fig. 2.1 Types of university contributions to the innovation endeavor. Source: Adapted from Ulrichsen and Kelleher (2022)

require major system transformations in how they are organized and the capabilities the need in place to deliver impact (Bodley-Scott & Oymak, 2022; Kelleher, 2023).

2.2.5 Growing Division of Labor in the Production, Translation, and Commercialization of Knowledge

These changes have, among other things, led to a growing division of labor in production, translation, and commercialization of knowledge within innovation systems (Arora et al., 2020). During the latter part of the twentieth century, research, particularly more fundamental, basic research, became increasingly dominated by universities and publicly funded research institutions; small startup companies—enabled by a growing venture capital market, became increasingly important in commercializing high-risk, promising inventions; and larger, more established companies focusing more on product development, systems integration, and market deployment (Arora et al., 2020; Arora & Gambardella, 1994).

This increasing division of labor has, however, created additional challenges for the innovation endeavor. Arora et al. (2020) raise concerns that we have yet to see the expected increase in productivity in translating scientific discoveries into economic wealth and competitive advantage. They suggest that the universities—through their technology transfer activities and partnerships with industry—startups and others, while delivering some major innovation successes, have failed to fully fill the gap left by the loss of major corporate R&D labs. These labs, they argue, were able to more easily assemble expertise across multiple knowledge domains, facilities, and other resources and integrate and direct them towards solving specific problems in ways that the more fragmented system struggles to resolve.

2.3 Changing Scale and Nature of UI Interaction

The nature and scale of interactions formed between universities and industry to drive and support innovation have co-evolved with the major shifts experienced by the national innovation systems in countries like the USA and UK since the Second World War. Compared with the period from 1945 to the 1970s, the attention by governments and industry to the contributions of universities to technology development, innovation, and national competitiveness was much deeper from the 1980s onwards, with greater attention to interaction between universities and industrial partners.

Research going back to the 1990s recognized the many pathways through which universities and companies interact to support innovation (Bonaccorsi & Piccaluga, 1994; Bozeman, 2000; Cohen et al., 2002; D’Este & Patel, 2007; Hughes & Kitson, 2012; Lester, 2005; Perkmann & Walsh, 2007; UIDP, 2012). Stretching well beyond creating new ventures to commercialize IP and technology licensing, these studies typically highlight a wide variety of mechanisms, including contract research, research partnerships, graduate fellowships, consultancy, prototype development and testing services, co-developing physical facilities, training, and the movement of people, for example through staff internships, joint appointments, secondments, and other forms of people exchange. They also highlight the role of informal interactions in facilitating knowledge transfer and exchange between the university base and industry, which in turn emphasizes the importance of activities that help to increase contact and build social relationships, such as networking, attending conferences involving both academia and industry, inviting industry onto campus, and giving lectures.

While universities and industrial partners have long interacted through a wide variety of mechanisms, we have witnessed long-term shifts since the 1980s that are worth noting. First, the scale of these links has increased in many countries. For example, in the US industry funded research in universities has grown from around

USD 2.2 billion in 1990 (in constant 2022 prices) to USD 5.7 billion in 2022.² In the UK, income to universities from a range of knowledge transfer activities (including not just industry funded research but consultancy, facilities hire, IP exploitation etc.) has grown from USD 4.3 billion in 2004 to USD 8.7 billion in 2022.³

Second, relational forms of interaction appear to have become much more important than more transaction-based interactions, reflecting not least the importance of tacit knowledge in the transfer of ideas and technologies between universities and industry (Schartinger et al., 2002).

Third, we have seen changes to the ways in which large companies organize their interactions with universities. The 2000s saw a trend from large numbers of ad hoc interactions with academics across many universities to a much smaller number of more strategically selected university partners (Geiger, 2010; Lambert, 2003). While having a long history in universities, such as MIT (Urban et al., 2003), these longer-term, more strategic partnerships began to proliferate in the 2010s and became much more important for both universities and companies alike (Bercovitz & Feldman, 2007; Ulrichsen & O'Sullivan, 2015).

UI consortia and center programs, which in the USA and UK became more prevalent following the introduction of dedicated funding programs in the 1980s to support their operation, continue to provide important vehicles for bringing partners together to work collectively on common research and innovation challenges. This is particularly the case when focused on pre-competitive, system-wide problems. However, efforts are needed to ensure these models remain effective moving forward, because they are increasingly used to organize efforts to tackle complex industrial and societal challenges (Ulrichsen, 2019).

Related to this has been a growing understanding of the different levels and models of interactions and when they should be used. For example, work by the UIDP (2012) highlights important differences between tactical and transactional interactions, collaborations, and strategic partnerships. Perkmann and Salter (2012) examine the value propositions for four types of collaboration, including idea labs for attracting new partners, building relationships and generating options; collaborations for tackling grand challenges that benefit from being structured through UI consortia, centers or other special-purpose vehicles; extended workbenches for solving near-term problems and accessing advice and support; and deep exploration models for tackling more fundamental challenges, which can benefit from long-term industry sponsorship of university-based research centers.

²US data based on National Center for Science and Engineering Statistics (NCSES). 2023. *Higher Education Research and Development: Fiscal Year 2022*. NSF 24–308. Alexandria, VA: National Science Foundation. Available at <https://ncses.nsf.gov/surveys/higher-education-research-development/2022>. Adjusted for US inflation using the US GDP deflator.

³UK data based on Higher Education Statistics Agency Higher Education Business and the Community (HEBCI) surveys 2003/04 –2021/22, available at <https://www.hesa.ac.uk/data-and-analysis>. Adjusted for UK inflation using the UK GDP deflator, and converted to US Dollars using purchasing power parity exchange rates available from the Organisation for Economic Cooperation and Development (OECD) data explorer.

Fourth, there appears to be greater importance placed on people exchange between universities and industry today. Arora et al. (2020) provide compelling examples of how, in the USA, the movement of scientists between academia and large corporate R&D labs, such as Xerox PARC and Bell Labs, was important in facilitating companies to develop and commercialize key technologies emerging from research. However, the analysis by Cohen et al. (2002) of a 1994 survey of R&D managers in US companies showed that just 5.8% of the 1478 respondents perceived people exchange to be an important pathway for supporting their R&D efforts. By contrast, this topic is often central to many discussions about how to strengthen the value and effectiveness of UI partnerships (e.g., NCUB, 2023, from the UK).

2.4 Options, Solutions, and Impact

Looking forward, recent summits bringing together senior leaders from leading universities, corporations, and government agencies from the USA, UK, and Europe have identified more trends and priorities as we look to the future of UI partnerships (Ulrichsen, 2019, 2023a). These include, as illustrated in Fig. 2.2:

- **Greater strategic focus and investment in local innovation systems** to underpin more effective partnerships, recognizing that increasingly complex challenges require a range of partners, infrastructure, talent, funding, and other resources.
- **Connect innovation partners:** Universities play important “public space” roles, providing an effective venue to bring people together to talk about common challenges, experiment with novel ways of solving problems, and catalyze interactions that may not otherwise form.
- **Broaden partnerships beyond traditional science and technology** to integrate the social sciences and humanities, recognizing that many complex problems require not just technological advances but advances elsewhere, such as behaviors, business models, working practices, ethics, regulations, and policy.
- **Experiment with innovative spaces and organizational “vehicles”** to facilitate collaborations that accelerate the translation of research into applications in ways that would be difficult through traditional discipline-focused departmental structures, for example by better enabling translational R&D; creating incentives and career pathways that attract wider sets of expertise; building physical and data infrastructure; and managing IP in different ways.
- **Securely share real-world resources**, such as industry-standard testing equipment, materials, labs, and datasets in ways that protect privacy, confidentiality and other sensitivities to underpin and unlock new areas of research and innovation.
- **Expand partnerships** to drive innovation in industries in need of transformation, such as construction, healthcare, legacy manufacturing industries, finance and professional services, public sector service delivery, and move beyond leading-edge corporates to work further down supply chains.

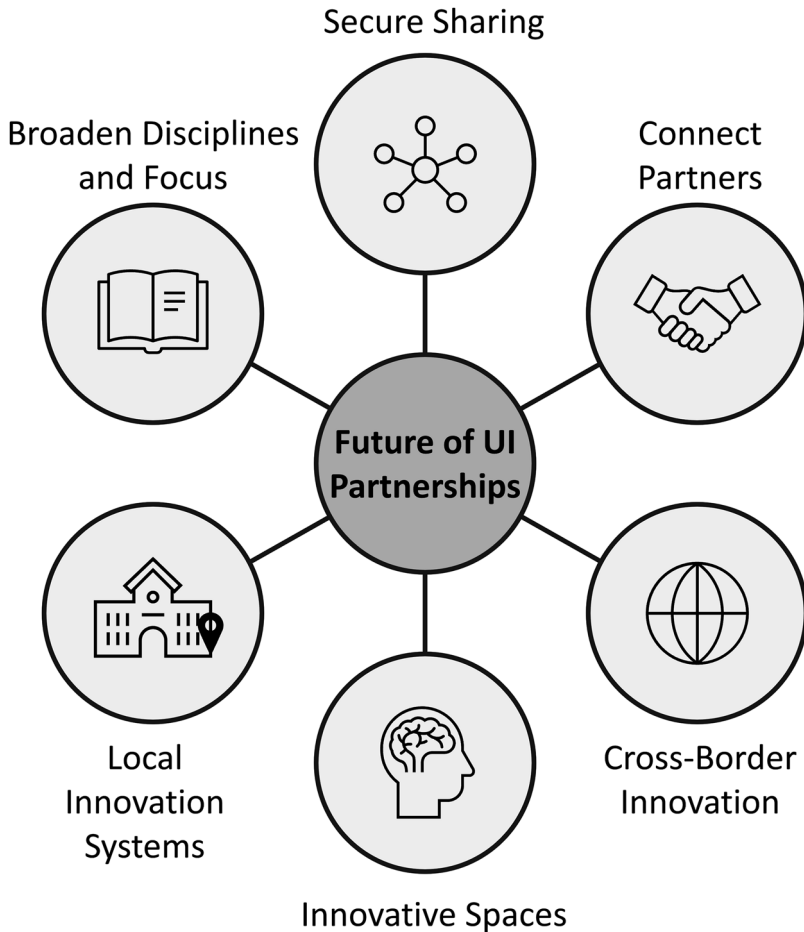


Fig. 2.2 Future of UI partnerships

- **Strengthen incentives** for individuals within academia and industry to collaborate and work on common problems. Despite efforts, current incentives still drive people to work within silos and hamper interdisciplinarity, cross-organization, and cross-border collaborations.
- **Innovate in how partners can work effectively across borders**, even amongst allies, given rising geopolitical tensions and concerns over security.

Finally, pressure from governments and societies is growing for research and innovation systems to address major grand societal challenges and crises, not least climate change, biodiversity loss, aging populations, pandemics, antimicrobial resistance, and resource scarcity. Demands are also growing for rejuvenating economies and turbocharging global competitiveness in the USA, UK, and across Europe. The research and innovation base is seen as a critical component of the system

needed to tackle these many and varied challenges. However, there are concerns that progress and solutions are not arriving fast enough or at sufficient scale. As a result, we are seeing new agencies and funding programs emerge, many of which are calling for partnership working across research and innovation systems, including universities, companies, technology development organizations, investors, governments, and societies. This approach recognizes that progress requires many partners to come together, often demanding transformation of entire systems.

A recent summit hosted at the University of Cambridge in 2024 examined how universities and their partners could better mobilize around tackling these types of complex grand challenges. It brought together senior leaders from across the research and innovation ecosystems from the USA, UK, and Europe. Through intensive discussions the summit highlighted that mobilizing around grand challenges entails:

- **Leadership at the system level** by people who can motivate, communicate, and build coalitions across diverse groups of stakeholders and communities.
- **Vision and alignment around common goals**, which must recognize that motivations for getting involved will vary across groups.
- **Communication** to get stakeholders and communities engaged and to get public buy-in. Clarity on the problem and approach is needed, and storytelling can be an important tool.
- **Internalizing the urgency and scale of the problem and the costs of inaction:** Politicians, industry, and society often struggle to grasp the urgency and long-term costs of grand challenges, especially when threats aren't immediate. This necessitates strategies to bridge the significant gap between investment and tangible results.
- **Understanding different types of grand challenges**, ranging from well-defined problems with largely technology-driven solutions, to much more open-ended crises requiring both technological advances and transformations in societies and economies.
- **Reducing the activation energy to engagement and providing a tractable pathway to solutions:** Even for the more complex challenges, there is significant merit in bringing people together to find ways of breaking these “wicked” problems into tamer, more tractable subproblems that people can better internalize and solve.

The summit also highlighted that complex grand challenges can benefit from partnerships that are:

- **Solutions-focused and multidimensional**, moving beyond individual activities to developing approaches that bring together different disciplines, integrate research, solutions development, and workforce development, and tackle system-wide changes that are needed to enable widespread deployment and adoption (e.g., changes to policies, behaviors, solution legitimacy, and payer pathways).
- **Designed for translation and scale-up** from the outset that anticipate the capabilities and resources required along the journey.

- **Aligned with stakeholder interest**, not least to provide clarity of purpose as well as of roles and expectations. Roadmapping tools can be valuable albeit costly.
- **Aimed at overcoming barriers to partnership**, for example around trust, communications, expectations management, working relationships, contracting, senior-level commitment, and sustaining interest beyond the initial period.
- **Better at integrating the knowledge value chain** that characterizes academia with the innovation value chains of industry, where handovers between them can be difficult and inefficient.
- **Capable of managing portfolios of projects** across multiple partners and sites, recognizing that progressing solutions will require greater coordination and coherence of activities than in other forms of partnership.
- **Agile**, ensuring that they can adapt to changing needs as progress is made towards solutions.

Making UI interactions work well is challenging, not least because it involves organizations coming together from different parts of the innovation system, which are governed by their own motivations, incentives, and objectives. These interactions are also shaped by the institutions, conditions, and dynamics of the innovation systems within which they exist (Ulrichsen, 2023b).

Many studies have sought to unearth factors that influence the process. Some examine specific forms of interactions—for example R&D partnerships (Bruneel et al., 2010; Mora-Valentin et al., 2004; Ulrichsen & O’Sullivan, 2015), UI centers (Feller et al., 2002), and spinouts (Siegel et al., 2003)—while others are more general (Hughes et al., 2016, 2022; Muscio & Vallanti, 2014). Furthermore, it is clear that some factors are structural (e.g., the gender or age of academics) while other factors are related to the degree to which something is present or absent (e.g., trust, communication, alignment). As a result of the latter, different studies have variously explored the same factor as a barrier (the lack of the factor) or an enabler (the presence of the factor), making comparability across the literature difficult.

Accepting these issues, Table 2.1 brings together what we know about the many factors that influence the interactions between universities and industrial partners. It crucially situates the interactions within the system of innovation and the wider organizational contexts of actors involved and the institutions (“rules of the game”) that are known to shape behaviors and actions. Despite these challenges, a large-scale survey of UK companies interacting with universities published in 2022 showed that most see these engagements as meeting or exceeding their expectations (Hughes et al., 2022).

2.5 Takeaways and Future Strategies

Universities and companies have interacted with each other to drive innovation for hundreds of years. Indeed, some of the most prominent research universities today were founded in the 1800s with a mission to address the needs of industry and their

Table 2.1 Factors influencing UI interactions

Area of system	Factors influencing interactions
University and department leadership	• Organization-level strategies and policies, incentives, and culture towards UI interactions • Past experiences, structures, and processes • Internal and external contexts • Capabilities of leadership to identify emerging opportunities and steer organization to seize them • Leadership commitment to UI interactions • Availability of resources and support structures to support and deliver interactions • Willingness and ability to experiment with new ways of working and mainstream successful initiatives • Suitability and effectiveness of administrative processes for engaging with partners
Academics	• Types of research (e.g., interdisciplinary, basic/use-inspired/applied) • Structural characteristics of individuals (gender, ethnicity, etc.) • Role seniority, scientific productivity, reputation • Identities (who I am) and motivations (is it worth it) towards engagement • Understanding of own local environment and constraints • Capacity and capability to engage (shaped by past experiences, training, networks, time availability & other commitments) • Access to project-specific resources for interactions • Awareness of opportunities and ability to identify partners
Interactions and projects	• Goal consensus/strategic fit and alignment and management of expectations • Trust, communication, and mutual understanding of needs and roles • Compatibility of cultures and working environments/practices • Appropriateness and diversity of interaction mechanisms • Relationship and project management and support • Flexibility and adaptability of relationships • Transactional costs of engagement (e.g., bureaucracy, negotiations, contracts) • Geographic proximity • Access to sufficient project-specific resources and support
Industry partners	• Partner strategic objectives and ambitions • Partner absorptive capacity (including capacity and capability to work collaboratively and to exploit knowledge generated with/by universities) • Partner capacities and capabilities to innovate • Stability of staff dedicated to working with universities • Availability of resources to invest in university engagements • Awareness of benefits of working with universities and incentives to engage • Ability to identify suitable university partners • Suitability and effectiveness of administrative processes for engaging universities

(continued)

Table 2.1 (continued)

Area of system	Factors influencing interactions
Institutions	National, regional, sector-specific institutions that shape choices and behaviors, e.g.: • Legal and IP frameworks • Financial system • Labor markets • Tax incentives • Government policies • Regulations and standards • Cultural norms and practices
Supply conditions	Availability of/access to: • Funding (public/private/charitable) • Innovation and entrepreneurial infrastructure and facilities • Skills/expertise (commercial, entrepreneurial, technical, managerial) • Networks • Complementary resources (knowledge, skills etc.) for interactions opportunity to be successful • Public spaces to facilitate connections and discussions
Demand conditions	• Sector structure, maturity, and dynamics • Innovativeness of sector and barriers to innovation • Sector capabilities to innovate, collaborate, and utilize knowledge from universities

Source: Adapted from Ulrichsen et al. (2023)

local economies, with effective UIRs central to success. That said, over the past 50 years and set against a backdrop of the decline of large corporate R&D labs, increasingly open and collaborative innovation processes, significant increases in public investment in R&D, and growing incentives for universities to translate and commercialize their research, we have witnessed important long-term shifts:

- The growing importance of UIRs for driving and enabling innovation in the economy
- The increasing scale and scope of UIRs, with relation-based interactions becoming much more important.
- The move by large companies from having many ad hoc interactions to more strategically selected university partners.
- The growing institutionalization of UIRs within formal university innovation and economic development missions.
- The investment by universities in systems of support for UIRs, including new or strengthened leadership roles, incentives, support offices and resources, engagement process, and the capabilities of academics and support staff to engage and deliver. In the UK this has been enabled by long-term flexible and strategic funding provided to the university.
- Growing pressures from governments and societies for universities, companies, investors and others to work together to address major, complex global and national challenges at pace, while also working to rejuvenate economies, turbo-charge competitiveness, and protect national security.

We also know that delivering effective UIRs is challenging, with many barriers hampering the process. Looking forward, given their importance to innovation, it is imperative that we find ways of making them work better and delivering greater value faster. In particular, we must:

- Create new incentives within universities and across the innovation system to increase engagement and to attract and retain the necessary expertise—including beyond the traditional academic—to accelerate progress towards innovative solutions.
- Better integrate research, solutions development, and workforce development around tackling major societal, national, and industrial challenges at pace and scale.
- Experiment with new organizational spaces better able to assemble the necessary partners, resources and expertise to accelerate the delivery of solutions to major challenges, and better manage the journey.
- Devote more effort to tackling and reducing long-standing frictions hampering the process.
- Invest in building local innovation ecosystems that put in place key building blocks for enabling partners to come together and accelerate the development, translation, and application of knowledge for societal gain.
- Broaden UIRs beyond traditional sectors and partners and ensure they help to drive not just industries of the future, but also help vital, traditional industries such as construction, healthcare, and agriculture to transform.
- Broaden the scope of UIRs to better integrate insights and expertise from the social sciences, arts, and humanities, recognizing that unleashing significant value through innovation often requires not just technological breakthroughs but significant changes to the way people work and behave and systems function.
- Demonstrate why and how universities and their partnerships with industry and others are important and deliver value to the world.

References

- Agar, J. (2019). *Science policy under Thatcher*. UCL Press. <https://doi.org/10.14324/111.9781787353411>
- Arora, A., & Gambardella, A. (1994). Evaluating technological information and utilizing it: Scientific knowledge, technological capability, and external linkages in biotechnology. *Journal of Economic Behavior and Organization*, 24(1), 91–114. [https://doi.org/10.1016/0167-2681\(94\)90055-8](https://doi.org/10.1016/0167-2681(94)90055-8)
- Arora, A., Belenzon, S., Pataconi, A., & Suh, J. (2020). The changing structure of American innovation: Some cautionary remarks for economic growth. *Innovation Policy and the Economy*, 20, 39–93. <https://doi.org/10.1086/705638>
- Arrow, K. (1962). *Economic welfare and the allocation of resources for invention [NBER Chapters]* (pp. 609–626). National Bureau of Economic Research. <http://econpapers.repec.org/bookchap/nbrmberch/2144.htm>

- Audretsch, D. B. (2014). From the entrepreneurial university to the university for the entrepreneurial society. *The Journal of Technology Transfer*, 39(3), 313–321. <https://doi.org/10.1007/s10961-012-9288-1>
- Bercovitz, J. E. L., & Feldman, M. P. (2007). Fishing upstream: Firm innovation strategy and university research alliances. *Research Policy*, 36(7), 930–948. <https://doi.org/10.1016/j.respol.2007.03.002>
- Bettis, R. A., & Hitt, M. A. (1995). The new competitive landscape. *Strategic Management Journal*, 16, 7–19.
- Bodley-Scott, T., & Oymak, E. (2022). *University–industry partnerships for positive change: Transformational strategic alliances towards UN SDGs* (1st ed.). Policy Press.
- Bonaccorsi, A., & Piccaluga, A. (1994). A theoretical framework for the evaluation of university–industry relationships. *R&D Management*, 24(3), 229–247. <https://doi.org/10.1111/j.1467-9310.1994.tb00876.x>
- Bozeman, B. (2000). Technology transfer and public policy: A review of research and theory. *Research Policy*, 29(4), 627–655.
- British Academy. (2019). *Lessons from the history of UK science policy*. The British Academy and the UK Department for Business, Enterprise and Industrial Strategy. <https://www.thebritishacademy.ac.uk/documents/243/Lessons-History-UK-science-policy.pdf>
- Bruneel, J., D'Este, P., & Salter, A. (2010). Investigating the factors that diminish the barriers to university–industry collaboration. *Research Policy*, 39(7), 858–868. <https://doi.org/10.1016/j.respol.2010.03.006>
- Caraga, J., Lundvall, B.-Å., & Mendonça, S. (2009). The changing role of science in the innovation process: From Queen to Cinderella? *Technological Forecasting and Social Change*, 76(6), 861–867. <https://doi.org/10.1016/j.techfore.2008.08.003>
- Chaminade, C., Lundvall, B. Å., & Haneef, S. (2018). Innovation policy and national innovation systems. In *Advanced introduction to national innovation systems* (pp. 107–114). Edward Elgar.
- Chen, A., Patton, D., & Kenney, M. (2016). University technology transfer in China: A literature review and taxonomy. *The Journal of Technology Transfer*, 41(5), 891–929. <https://doi.org/10.1007/s10961-016-9487-2>
- Chesbrough, H. W. (2003). *Open innovation: The new imperative for creating and profiting from technology*. Harvard Business School Press.
- Cohen, W. M., Nelson, R. R., & Walsh, J. P. (2002). Links and impacts: The influence of public research on industrial R&D. *Management Science*, 48(1), 1–23.
- Conn, R., Crow, M., Friend, C., & McNutt, M. (2021). *The next 75 years of U.S. science and innovation policy: An introduction*.
- D'Este, P., & Patel, P. (2007). University–industry linkages in the UK: What are the factors underlying the variety of interactions with industry? *Research Policy*, 36(9), 1295–1313.
- Edquist, C. (1997). *Systems of innovation: Technologies, institutions and organizations*. Routledge.
- Feller, I., Ailes, C. P., & Roessner, J. D. (2002). Impacts of research universities on technological innovation in industry: Evidence from engineering research centers. *Research Policy*, 31(3), 457–474.
- Foray, D. (2004). *Economics of knowledge*. MIT Press.
- Freeman, C. (1982). *The economics of industrial innovation*. Pinter.
- Galanakis, K. (2006). Innovation process. Make sense using systems thinking. *Technovation*, 26(11), 1222–1232.
- Geiger, R. L. (2010). University supply and corporate demand for academic research. *The Journal of Technology Transfer*, 37(2), 175–191. <https://doi.org/10.1007/s10961-010-9192-5>
- Gerybadze, A., & Reger, G. (1999). Globalization of R&D: Recent changes in the management of innovation in transnational corporations. *Research Policy*, 28(2–3), 251–274. [https://doi.org/10.1016/S0048-7333\(98\)00111-5](https://doi.org/10.1016/S0048-7333(98)00111-5)
- HMSO. (1993). *Realising our potential: A strategy for science, engineering and technology* (Cm 2250). Her Majesty's Stationery Office.

- Hughes, A., & Kitson, M. (2012). Pathways to impact and the strategic role of universities: New evidence on the breadth and depth of university knowledge exchange in the UK and the factors constraining its development. *Cambridge Journal of Economics*, 36(3), 723–750.
- Hughes, A., Lawson, C., Kitson, M., & Salter, A. (2016). *The changing state of knowledge exchange: UK academic interactions with external Organisations 2005–2015*. National Centre for Universities and Business.
- Hughes, A., Kitson, M., Salter, A., Angenendt, D., & Hughes, R. (2022). *The changing state of business-university interactions in the UK 2005 to 2021* (p. 120). National Centre for Universities and Business and University of Cambridge Centre for Business Research.
- Kelleher, L. (2023). *University models for sustainable development: Clarifying a complex landscape (02; UCI expert insights report)*. Policy Evidence Unit for University Commercialisation and Innovation, University of Cambridge.
- Kelleher, L., & Ulrichsen, T. C. (2022). *Innovation policy and the university system in times of crisis and recovery: What we know and what it means*. Policy Evidence Unit for University Commercialisation and Innovation, University of Cambridge.
- Kline, S. J., & Rosenberg, N. (1986). An overview of innovation. In National Research Council (Ed.), *The positive sum strategy: Harnessing technology for economic growth*. National Academies Press.
- Kuznets, S. (1971). *Economic growth of nations: Total output and production structure*. Belknap Press of Harvard University Press.
- Lambert, R. (2003). *Lambert review of business-university collaboration: Final report*. HM Treasury.
- Laursen, K., & Salter, A. (2006). Open for innovation: The role of openness in explaining innovation performance among U.K. manufacturing firms. *Strategic Management Journal*, 27(2), 131–150. <https://doi.org/10.1002/smj.507>
- Lester, R. (2005). *Universities, innovation, and the competitiveness of local economies. A summary report from the local innovation systems project: Phase I*. Massachusetts Institute of Technology, Industrial Performance Center, Working Paper Series. <http://web.mit.edu/lis/papers/LIS05-010.pdf>
- Lindner, R., Edler, J., & Daimer, S. (2024). Understanding paradigm change in science, technology, and innovation policy: Between science push and policy pull. In J. Edler & R. Walz (Eds.), *Systems and innovation research in transition research questions and trends in historical perspective*. Springer Nature Switzerland.
- Lundvall, B.-A., Johnson, B., Andersen, E. S., & Dalum, B. (2002). National systems of production, innovation and competence building. *Research Policy*, 31(2), 213–231.
- Martin, B. R. (2012). Are universities and university research under threat? Towards an evolutionary model of university speciation. *Cambridge Journal of Economics*, 36(3), 543–565. <https://doi.org/10.1093/cje/bes006>
- Mazzucato, M. (2018). Mission-oriented innovation policies: Challenges and opportunities. *Industrial and Corporate Change*, 27(5), 803–815. <https://doi.org/10.1093/icc/dty034>
- Mora-Valentin, E. M., Montoro-Sanchez, A., & Guerras-Martin, L. A. (2004). Determining factors in the success of R&D cooperative agreements between firms and research organizations. *Research Policy*, 33(1), 17–40. [https://doi.org/10.1016/S0048-7333\(03\)00087-8](https://doi.org/10.1016/S0048-7333(03)00087-8)
- Muscio, A., & Vallanti, G. (2014). Perceived obstacles to university–industry collaboration: Results from a qualitative survey of Italian academic departments. *Industry and Innovation*, 21(5), 410–429. <https://doi.org/10.1080/13662716.2014.969935>
- NCUB. (2023). *Pathways to success: NCUB researcher career mobility taskforce 2023*. National Centre for Universities and Business. <https://www.ncub.co.uk/wp-content/uploads/2021/07/NCUBs-Pathway-to-Success-web.pdf>
- PACEC/CBR. (2009). *Evaluation of the effectiveness and role of HEFCE/OSI third stream funding* (Issues Paper 2009/15, p. 240). HEFCE.
- Perkmann, M., & Salter, A. (2012). How to create productive partnerships with universities. *MIT Sloan Management Review*, 53, 79–88.

- Perkmann, M., & Walsh, K. (2007). University–industry relationships and open innovation: Towards a research agenda. *International Journal of Management Reviews*, 9(4), 259–280. <https://doi.org/10.1111/j.1468-2370.2007.00225.x>
- Santoro, M. D., & Chakrabarti, A. K. (2001). Corporate strategic objectives for establishing relationships with university research centers. *IEEE Transactions on Engineering Management*, 48(2), 157–163.
- Schartinger, D., Rammer, C., Fischer, M. M., & Fröhlich, J. (2002). Knowledge interactions between universities and industry in Austria: Sectoral patterns and determinants. *Research Policy*, 31, 303–328.
- Schot, J., & Steinmueller, W. E. (2018). Three frames for innovation policy: R&D, systems of innovation and transformative change. *Research Policy*, 47(9), 1554–1567. <https://doi.org/10.1016/j.respol.2018.08.011>
- Shane, S. (2004). Encouraging university entrepreneurship? The effect of the Bayh-Dole Act on university patenting in the United States. *Journal of Business Venturing*, 19(1), 127–151. [https://doi.org/10.1016/S0883-9026\(02\)00114-3](https://doi.org/10.1016/S0883-9026(02)00114-3)
- Siegel, D. S., Waldman, D. A., Atwater, L. E., & Link, A. N. (2003). Commercial knowledge transfers from universities to firms: Improving the effectiveness of university–industry collaboration. *The Journal of High Technology Management Research*, 14(1), 111–133. [https://doi.org/10.1016/S1047-8310\(03\)00007-5](https://doi.org/10.1016/S1047-8310(03)00007-5)
- Solow, R. M. (1957). Technical change and the aggregate production function. *The Review of Economics and Statistics*, 39(3), 312–320. <https://doi.org/10.2307/1926047>
- Stokes, D. E. (1997). *Pasteur's quadrant: Basic science and technological innovation*. Brookings Institution Press.
- Tassey, G. (2007). *The technology imperative*. Edward Elgar.
- Tidd, J. (2006). *A review of innovation models*. http://www.emotools.com/static/upload/files/innovation_models.pdf
- UIDP. (2012). *Partnership continuum: Understanding and developing the pathways for beneficial university–industry engagement*. University-Industry Demonstration Partnership.
- Ulrichsen, T. C. (2015). *Assessing the economic impacts of the higher education innovation fund: A mixed-method quantitative assessment*. HEFCE.
- Ulrichsen, T. C. (2019). *Developing university–industry partnerships fit for the future [Report for Oxford-UIDP Summit 2019]* (p. 49). University of Oxford.
- Ulrichsen, T. C. (2023a). *Rising to the challenge: Mobilising university–industry–government partnerships to lead an innovation-led recovery [report for the Oxford summit 2021]*. University of Oxford.
- Ulrichsen, T. C. (2023b). *Universities at the heart of an innovation-led recovery: What we know about how universities contribute to innovation, the factors that influence the process, and their potential to drive the recovery (IRC insight paper 005)*. Innovation and Research Caucus.
- Ulrichsen, T. C. (2024). *Configuring for impact: Building systems of support for university–business knowledge exchange in the English higher education sector*. In K. Daniels & S. L. Hansen (Eds.), *How to enable engagement between universities and business: A guide for building relationships*. Edward Elgar. <https://www.elgaronline.com/edcollchap/book/9781035316809/book-part-9781035316809-20.xml>
- Ulrichsen, T. C., & Kelleher, L. (2022). *Through crisis to recovery: The ongoing effects of the COVID-19 pandemic on universities and their ability to drive innovation [UCI/NCUB Policy Report]* (p. 100). UCI Policy Evidence Unit and NCUB.
- Ulrichsen, T. C., & O'Sullivan, E. (2015). *Building long term strategic university–industry partnerships: Lessons and effective practices from UK and US experiences* (p. 52). Centre for Science, Technology and Innovation Policy, University of Cambridge.
- Ulrichsen, T. C., Kelleher, L., & Roupakia, Z. (2023). *Knowledge exchange funding review: Insights from a review of the literature*. Policy Evidence Unit for University Commercialisation and Innovation, University of Cambridge.

- Urban, G. L., Glicksman, L. R., Koster, K. F., Lees, J. A., & Roth, G. L. (2003). *MIT'S industrial partnerships* (pp. 1–37). Massachusetts Institute of Technology.
- Weber, K. M., & Rohracher, H. (2012). Legitimizing research, technology and innovation policies for transformative change. *Research Policy*, 41(6), 1037–1047. <https://doi.org/10.1016/j.respol.2011.10.015>
- Youtie, J., & Shapira, P. (2008). Building an innovation hub: A case study of the transformation of university roles in regional technological and economic development. *Research Policy*, 37(8), 1188–1204. <https://doi.org/10.1016/j.respol.2008.04.012>

Tomas Coates Ulrichsen is the Director of the Policy Evidence Unit for University Commercialisation and Innovation (UCI) at the University of Cambridge and National Advisor to the English universities funding agency, Research England, on knowledge exchange metrics. Over the past decade his work has been instrumental in informing the development of UK funding programs aimed at strengthening the contributions of universities to innovation and in providing insights on how to build and nurture more effective partnerships between universities, companies, and others to accelerate innovation.